

Technical Report

Investigating the Relationship Between Operational Performance and Customer Repeat Purchasing in an E-commerce Marketplace

A Diagnostic Analysis of First-Order Delivery Performance, Freight Burden, and Repeat Purchase Behaviour

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Program: Data Analytics — Bring Your Own Capstone (BYOC)

Dataset: Olist Brazilian E-Commerce Public Dataset (Kaggle)

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1. Executive Summary

Repeat purchase is the clearest signal of customer satisfaction and long-term value in an e-commerce marketplace, yet only 3.09% of customers in the Olist Brazilian E-Commerce dataset placed a second order. This project investigates whether two observable, first-order operational experiences — delivery lateness and shipping (freight) burden — are statistically associated with whether a customer returns to buy again, using a pre-registered, hypothesis-driven methodology on a customer-level analytical population of 93,254 customers.

The analysis finds a statistically significant, though modest, association between first-order delivery lateness and reduced repeat purchase (chi-square test, $p = 0.0059$; odds ratio = 0.811). No statistically significant association was found between first-order freight burden and repeat purchase once the freight metric was corrected for a mechanical confound with order value ($p = 0.968$). A formal interaction test further found no statistically significant difference in the delivery effect across customer-value tiers ($p = 0.341$). A held-out predictive baseline confirms these are individually weak effects: a logistic regression using the four first-order predictors achieves a test ROC-AUC of only 0.5466, barely above the 0.500 random-guess baseline.

The primary business recommendation is to prioritise first-order delivery reliability as the main retention-related operational focus, ranking customer segments by exposure (volume of late first orders) and customer value for resource allocation — not on the basis of a differential delivery effect, since none was found to be statistically significant. Freight burden is not recommended as a retention lever. Because the underlying analysis is observational, a controlled operational test is recommended before committing a large intervention budget.

A methodological correction is a deliberate and disclosed part of this project: an initial freight-to-price ratio metric showed a statistically significant effect, which on closer inspection was found to be a mechanical artifact of the ratio's confound with order value. Rebuilding the metric on absolute freight value, with order value held as a separate control, made the effect disappear entirely. Both versions are reported side by side in this document and in the underlying notebooks, rather than silently carrying forward the uncorrected result.

2. Business Problem and Objectives

E-commerce marketplaces depend on repeat customers for sustainable growth, since acquiring a new customer is materially more expensive than retaining an existing one. Within the Olist marketplace, only a small fraction of customers return for a second purchase. Marketplace operators typically have limited direct control over product assortment or pricing set by third-party sellers, but they do have significant influence over two operational levers: the reliability of order fulfilment and delivery, and the cost of shipping passed on to the customer.

This project asks a diagnostic, not merely descriptive, question: are these two operational levers — first-order delivery lateness and first-order freight burden — measurably associated with whether a customer returns to purchase again? The objective is to answer this question with statistical rigour (formal hypothesis tests, effect sizes, confidence intervals) rather than only descriptive comparison, and to translate the answer into prioritised, evidence-bounded business recommendations rather than speculative ones.

Research Questions

- RQ1: Is a late first-order delivery associated with a lower repeat-purchase rate?
- RQ2: Is a higher first-order freight burden associated with a lower repeat-purchase rate?
- RQ3: Does the delivery-lateness effect on repeat purchase differ across customer-value segments?
- RQ4: How much of the observed repeat-purchase variation can the first-order experience variables predict out-of-sample, and what does that imply for using them as a targeting rule?

3. Dataset and Data Sourcing

3.1 Source and Licence

The analysis uses the Olist Brazilian E-Commerce Public Dataset, published on Kaggle by Olist, a Brazilian e-commerce marketplace integrator. The dataset is released under a CC BY-NC-SA 4.0 licence (verified directly against the live Kaggle dataset page). It covers approximately 99,441 orders placed between September 2016 and October 2018, spread across nine relational CSV files covering orders, order items, payments, reviews, products, sellers, customers, geolocation, and product category translations. All customer and seller identifiers are anonymised by the original publisher; no personally identifiable information is present in the dataset.

3.2 Data Validation

Prior to analytical processing, all nine raw files were validated for completeness, key and relationship integrity, cardinality, and known data-quality issues. This validation confirmed: all nine expected files present; no missing or negative values in price, freight, or payment fields; timestamp fields require type coercion (via TRY_CAST) but are otherwise complete for 96,476 of 99,441 orders' actual delivery dates; and no duplicate order or customer identifiers after cleaning. This evidence trail is documented in Notebook 01 (Data Inventory and Validation).

3.3 Analytical Population Construction

The analytical grain for this project is one row per customer, using each customer's first order (by purchase timestamp) as the customer-level record. This choice is deliberate: an earlier order-level exploratory pass revealed a unit-of-analysis bias when comparing repeat and non-repeat customers using order-level records directly, since repeat customers contribute multiple rows and their later orders cannot influence a decision that has already been made. Using the first order for every customer restores temporal precedence and avoids this bias.

The customer-level analytical population is further restricted to customers whose first order has a valid actual delivery date, since delivery lateness cannot be measured for an order that was never delivered. This restriction is not neutral, and its effect is quantified and disclosed in Section 7 (Limitations).

Population	Customers	Repeat Customers	Repeat Rate
Retained (first order delivered) — used throughout this analysis	93,254	2,877	3.09%
Excluded (first order never delivered)	2,842	120	4.22%

Table 1. Analytical population, retained vs excluded (Notebook 02, Section 10a).

4. Methodology

4.1 Data Cleaning and Transformation

DuckDB is used as the primary analytical engine throughout this project, chosen for its in-process SQL execution and lazy evaluation over the multi-gigabyte join surface implied by combining all nine source tables. Raw CSV files are registered as DuckDB views and cleaned entirely in SQL: timestamp fields are converted with `TRY_CAST` (which yields `NULL` rather than raising an error on a malformed value), delivery duration and delay are computed directly in SQL, and one-to-many child tables (order items, payments, reviews) are aggregated to order grain before being joined to the orders table, preventing row duplication. Pandas is used only for display and for exporting convenience CSV copies of the DuckDB tables.

A customer's first order is identified using a `ROW_NUMBER()` window function partitioned by `customer_unique_id` and ordered by purchase timestamp; a `COUNT(*)` window function over the same partition gives each customer's total order count, which defines the `is_repeat_customer` flag (1 if the customer placed more than one order, 0 otherwise).

4.2 Feature Engineering

Feature engineering (Notebook 03) is limited to variables with clear analytical meaning for the stated hypotheses: a first-order delivery lateness flag, first-order freight burden (in two forms — see Section 5.3), first-order order value, customer value tiers (quartile-based on total first-order value, giving four approximately equal-sized groups of ~23,300 customers each), and winsorized (1st/99th percentile capped) variants of order value and freight for outlier-sensitivity checks.

4.3 Statistical Methodology

All formal hypotheses are pre-registered prior to testing, and results are reported regardless of direction or significance.

- H1 (delivery lateness): chi-square test of independence, with an odds ratio and 95% confidence interval for the effect size.
- H2 (freight burden): two-proportion z-test between low- and high-freight-burden groups (quartile bands), supplemented by a Mann-Whitney U test and Cliff's delta within order-value quintiles to control for the price-freight relationship.
- Multicollinearity check: Spearman correlation and Variance Inflation Factor (VIF), performed before fitting the confirmatory model, not after.
- Confirmatory model: logistic regression (statsmodels), reporting odds ratios and 95% confidence intervals, used for inference on the full population.
- Predictive baseline: the same specification fit with scikit-learn's LogisticRegression on a stratified 75/25 train/test split, evaluated using ROC-AUC and a confusion matrix rather than raw accuracy, given the ~3% positive-class rate.
- Interaction test: a likelihood-ratio test comparing a main-effects-only model against a model with late-flag x value-tier interaction terms, to formally test RQ3.

All tests use a significance threshold of $\alpha = 0.05$. Effect sizes and confidence intervals are reported alongside every p-value so that statistical significance is never presented without its corresponding practical magnitude.

5. Results

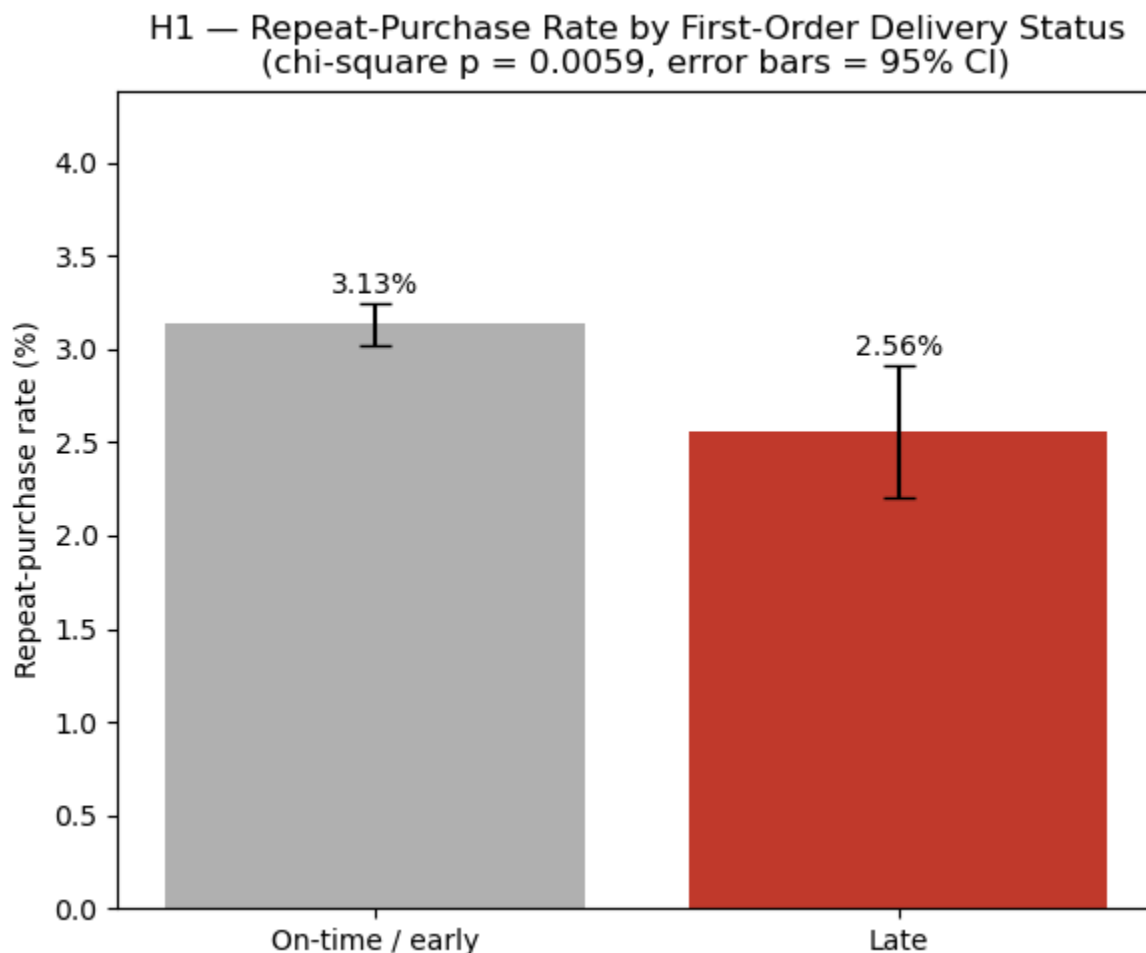
5.1 Hypothesis 1 — First-Order Delivery Lateness

H1: Customers whose first order was delivered late have a lower repeat-purchase rate than customers whose first order arrived on time or early.

Group	Customers	Repeat Customers	Repeat Rate
On-time / early first order	85,662	2,683	3.13%
Late first order	7,592	194	2.56%

Table 2. H1 descriptive comparison (Notebook 04 / Notebook 06).

The chi-square test of independence is statistically significant ($p = 0.0059$). The odds of repeat purchase for a customer with a late first order are 0.811 times the odds for a customer with an on-time or early first order (95% CI for the repeat-rate difference: -0.95 to -0.20 percentage points). This represents a real but modest effect: late delivery is associated with a 0.58 percentage-point reduction in repeat-purchase rate.



5.2 Hypothesis 2 — First-Order Freight Burden

H2: Higher first-order freight burden is associated with lower repeat purchase. This hypothesis required a methodological correction, described in the following section, before it could be answered reliably.

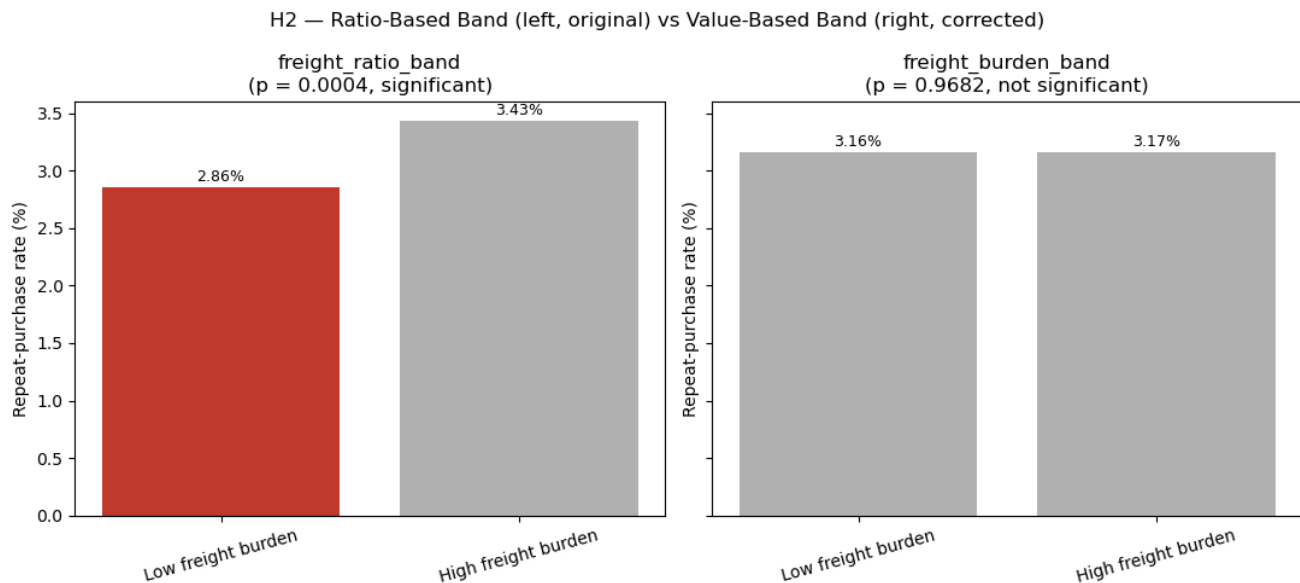
5.3 Methodological Correction — The Freight-Ratio Artifact

The freight burden metric was initially operationalised as a freight-to-price ratio (freight value divided by order value). This ratio was found to be mechanically confounded with order value itself: a cheap order with an entirely ordinary freight charge produces a high ratio, while an expensive order with a genuinely high freight charge can produce a low ratio. A Spearman correlation of freight value against order value of approximately 0.47 (well below the VIF-flagged threshold) confirmed the two are related but not collinear enough to explain this reversal on their own — the ratio construction itself was the source of the artifact.

Freight Metric	Low Band Repeat Rate	High Band Repeat Rate	Difference (pp)	p-value	Verdict
Ratio-based (original)	2.86%	3.43%	+0.58	0.0004	Significant
Value-based (corrected)	3.16%	3.17%	+0.01	0.968	Not significant

Table 3. H2 result under both freight-band constructions (Notebook 04, Section 11b). The ratio-based band's significant result does not survive correction to an absolute-value band.

The metric was rebuilt on absolute first-order freight value (with order value retained as a separate control variable, rather than removed, at the analyst's explicit direction during project design) and both versions are reported side by side rather than silently replacing one with the other. Under the corrected, value-based band, the repeat-rate difference between high- and low-freight-burden customers is 0.01 percentage points and is not statistically significant ($p = 0.968$). This confirms the original result was an artifact of the ratio's mechanical dependence on price, not a genuine relationship between freight and retention.



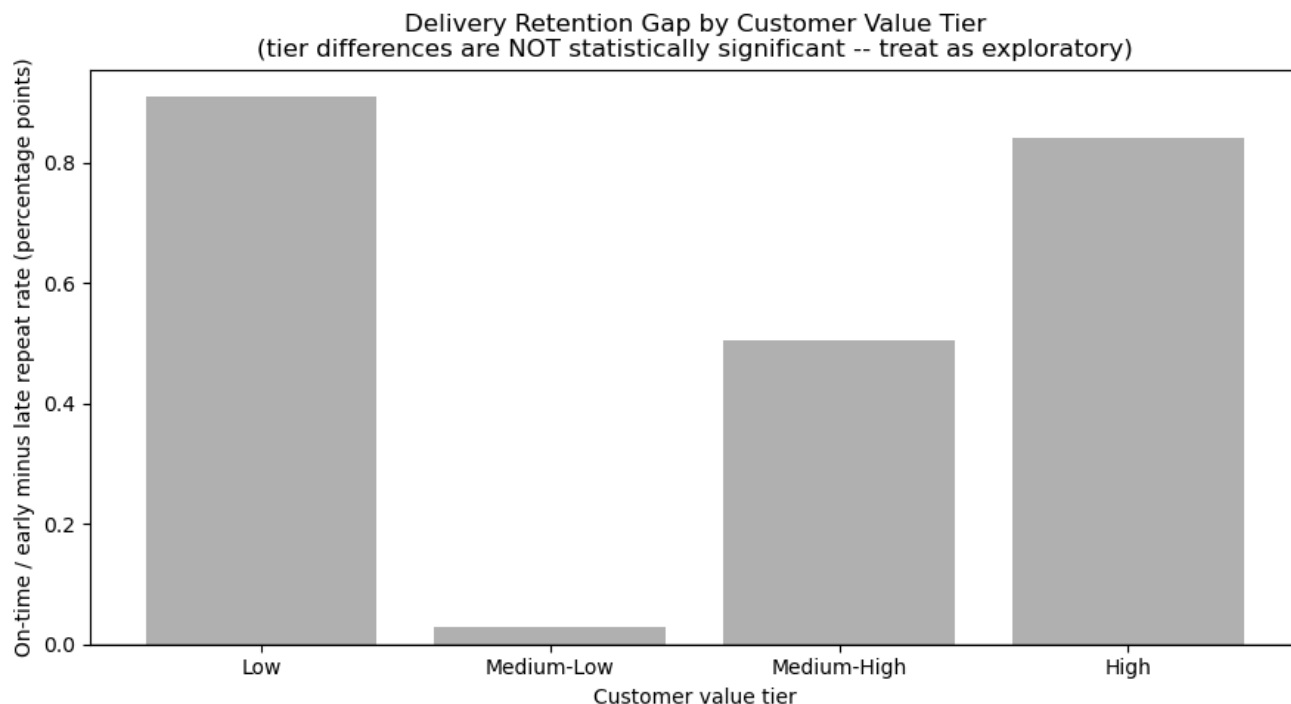
5.4 Delivery Effect Across Customer-Value Tiers

A formal likelihood-ratio test compares a logistic regression with late-flag x value-tier interaction terms against a main-effects-only model. The interaction is not statistically significant (likelihood-ratio statistic = 3.350, $df = 3$, $p = 0.341$).

Value Tier	On-Time Repeat Rate	Late Repeat Rate	Retention Gap (pp)
Low	3.29%	2.38%	0.91
High	3.02%	2.18%	0.84

Value Tier	On-Time Repeat Rate	Late Repeat Rate	Retention Gap (pp)
Medium-High	2.98%	2.47%	0.51
Medium-Low	3.24%	3.21%	0.03

Table 4. Descriptive delivery retention gap by customer-value tier (Notebook 05/06). The spread across tiers is not statistically distinguishable from sampling noise around one overall effect ($p = 0.341$).



This descriptive spread (0.03 to 0.91 percentage points) is consistent with sampling noise around a single overall delivery effect rather than genuine segment-level variation. The practical implication is that customer-value tiers can still be used to prioritise resource allocation (larger absolute opportunity in higher-value or higher-exposure segments), but should not be presented as evidence that the delivery problem itself is worse for any particular segment.

5.5 Confirmatory Logistic Regression

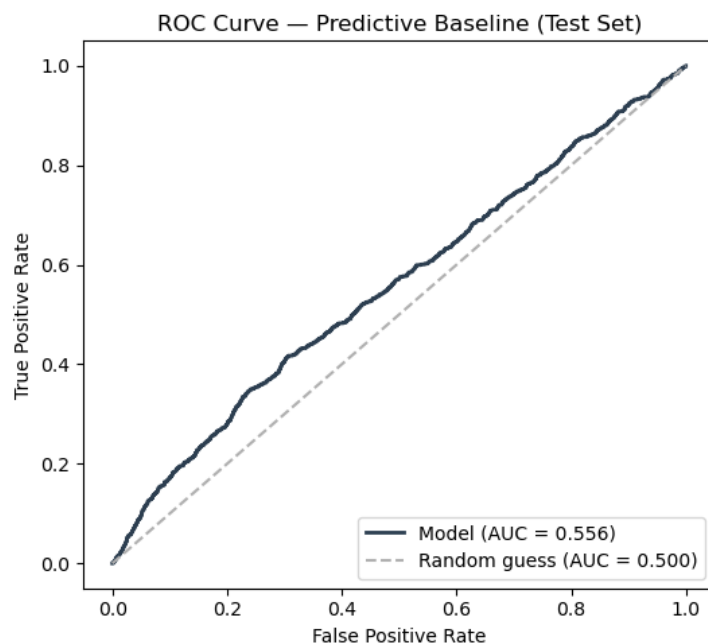
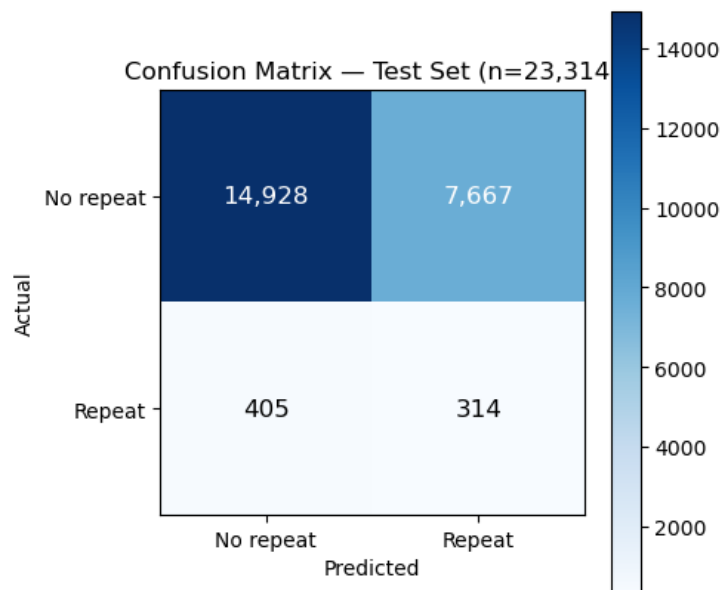
A multivariable logistic regression ($\text{is_repeat_customer} \sim \text{first_order_late_flag} + \text{first_order_freight} + \text{first_order_value} + \text{first_order_installments}$) was fit on the full analytical population of 93,254 customers, after confirming no material multicollinearity among predictors (all VIF values below 5). This model serves as the primary inferential result: it estimates the independent association of each first-order variable with repeat purchase while holding the others constant, and its odds ratio for first-order lateness is consistent in direction and magnitude with the standalone chi-square result in Section 5.1.

5.6 Predictive Baseline — Train/Test Evaluation

To evaluate whether the same first-order variables have practical predictive value for an individual customer (as opposed to a population-level association), the confirmatory specification was refit using scikit-learn's LogisticRegression with `class_weight="balanced"` on a stratified 75/25 train/test split (69,939 training customers, 23,314 test customers, repeat rate preserved at approximately 3.08-3.09% in both splits).

Metric	Value	Interpretation
Test accuracy	66.13%	Below the 96.92% trivial "always predict no-repeat" baseline; expected given class-balanced weighting, not a defect
Test ROC-AUC	0.5466	Weak — barely above the 0.500 random-guess baseline
Recall (repeat class)	39.2%	282 of 719 actual repeat customers correctly flagged
Precision (repeat class)	3.6%	282 correct flags among 7,741 total positive predictions

Table 5. Predictive baseline evaluation, test set (Notebook 04, Section 13a).



This result is consistent with, not contradictory to, the inferential findings above: H1's effect size (odds ratio 0.811) and H2's effect size (near-zero under the corrected banding) are both small at the population level, and a weak population-level association is expected to translate into weak individual-level predictive power. The practical conclusion is that first-order delivery lateness is a real, statistically defensible signal to act on operationally at the population level, but the four first-order variables tested here are not, on their own, a reliable tool for predicting or targeting individual customers.

6. Business Analysis and Recommendations

6.1 Segment Expected Repeat Revenue

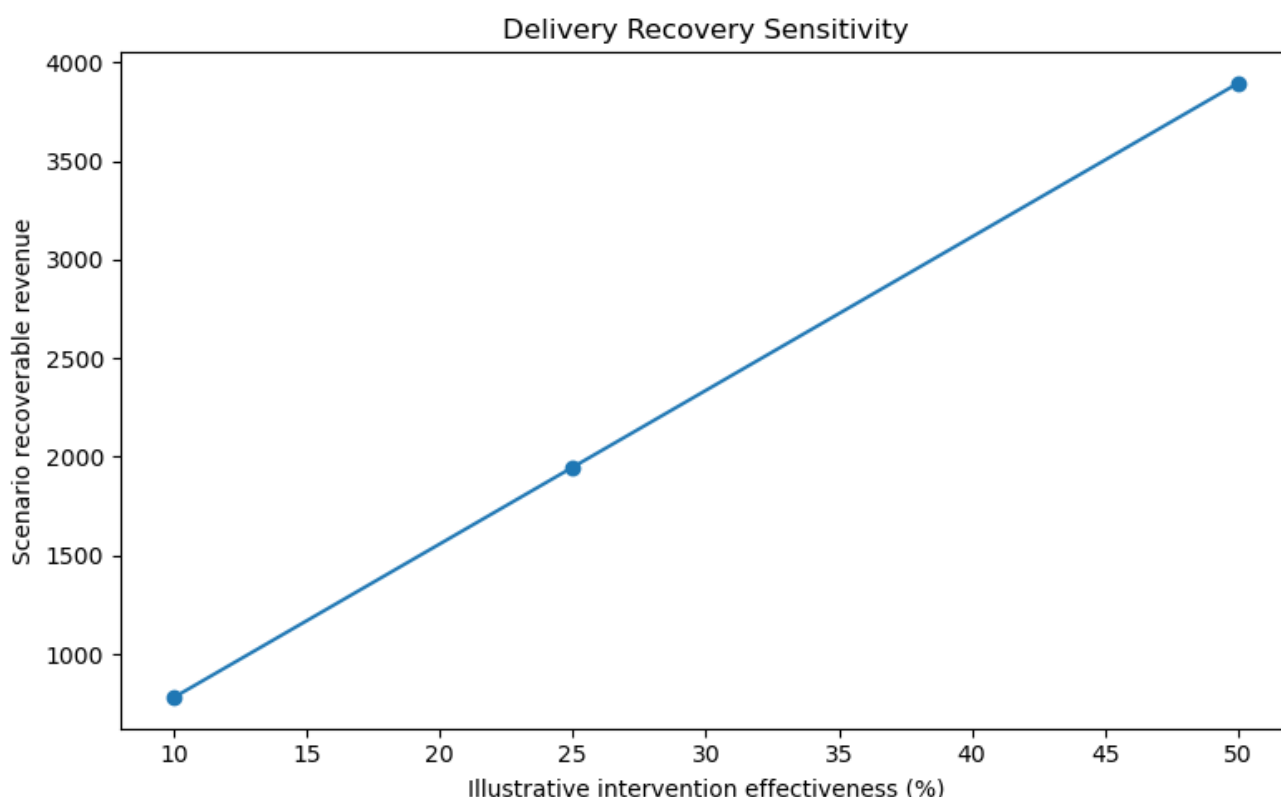
Segment Expected Repeat Revenue is calculated as average first-order value multiplied by the segment's historical repeat rate — a deliberately simple expected-value proxy for what one repeat purchase is worth on average within a segment. This is explicitly not a customer lifetime value or discounted-cash-flow model, and is not presented as one.

Value Tier	Customers	Avg Order Value	Repeat Rate	Segment Expected Repeat Revenue
Low	23,400	R\$28.61	3.22%	R\$0.92
Medium-Low	23,293	R\$63.94	3.24%	R\$2.07
Medium-High	23,401	R\$115.10	2.94%	R\$3.38
High	23,160	R\$344.04	2.94%	R\$10.13

Table 6. Segment Expected Repeat Revenue by customer-value tier (Notebook 05).

6.2 Delivery Opportunity and Sensitivity

A gross delivery opportunity figure (late customers x average order value x the positive portion of the observed retention gap) totals approximately R\$7,788.63 across all four value tiers. This is explicitly an illustrative upper bound, not a revenue forecast, since the dataset does not observe how effective any actual delivery-improvement intervention would be. Three illustrative intervention-effectiveness scenarios (10%, 25%, and 50%) are provided instead of a single invented recovery rate, to be replaced with business-approved estimates if and when they become available.



6.3 Recommendations

Recommendation 1 — Improve first-order delivery reliability

Prioritise first-order delivery reliability as the primary retention-related operational improvement area, supported by both the significant chi-square result (Section 5.1) and the consistent direction of the confirmatory logistic regression (Section 5.5).

Recommendation 2 — Prioritise by exposure and value, not by a differential effect

Rank customer segments for resource allocation using a combination of customer value, volume of late first orders, and expected repeat-purchase value. This is a resource-allocation choice, not a claim that the underlying delivery effect is stronger in any one segment — the formal interaction test (Section 5.4) found no statistically significant difference across value tiers.

Recommendation 3 — Do not treat freight as a retention lever

Do not recommend a freight-reduction intervention on retention grounds. The corrected, value-based freight analysis (Section 5.3) shows no statistically significant relationship with repeat purchase in either direction. Freight should continue to be monitored for margin and customer-experience purposes, but not presented as a demonstrated retention driver.

Recommendation 4 — Validate through a controlled operational test

Because the underlying analysis is observational, it establishes an association, not a causal mechanism. Before committing a large operational budget to delivery improvements, run a controlled test comparing customers who receive an improved delivery experience against comparable customers receiving the existing experience, with repeat purchase within a defined follow-up period as the success measure.

6.4 KPI Framework

KPI	Purpose	Frequency
First-order late-delivery rate	Monitor delivery performance	Weekly / monthly
Repeat-purchase rate (overall)	Track overall retention	Monthly
Repeat-purchase rate by delivery outcome	Track repeat behaviour by delivery experience	Monthly
Delivery retention gap	Monitor the observed delivery-retention difference	Monthly
Segment Expected Repeat Revenue	Estimate value represented by a repeat purchase	Monthly
Scenario recoverable revenue	Evaluate intervention scenarios	Scenario review

Table 7. Recommended KPI monitoring framework (Notebook 06).

7. Limitations

1. The dataset is observational; association does not establish causality.
2. The repeat-purchase definition is based on the analytical population constructed in the cleaning and feature-engineering stages, using each customer's first order.
3. The dataset does not directly observe customer motivations or intentions.
4. Intervention effectiveness is not available in the source data; revenue-opportunity figures are scenario estimates, not guaranteed forecasts.
5. The analysis does not estimate long-term customer lifetime value; Segment Expected Repeat Revenue is a single-repeat-purchase proxy only.
6. The analytical population excludes 2,842 customers whose first order was never delivered. This exclusion is necessary to measure delivery lateness, but is not neutral: the excluded population's repeat rate (4.22%) is significantly higher than the retained population's (3.09%; two-proportion z-test, $z = 3.436$, $p = 0.0006$). The exclusion is therefore conservative — it does not inflate the measured H1 effect; if anything, it removes a higher-retention group from the analysis.
7. The value-tier segment-priority recommendation (Recommendation 2) is a resource-allocation choice, not evidence of a differential effect: the formal interaction test found no statistically significant difference in the delivery effect across customer-value tiers ($p = 0.341$).
8. Freight burden did not show the expected negative retention pattern under the corrected (value-based) banding, so it is not presented as a proven retention problem in either direction.
9. The predictive baseline (ROC-AUC = 0.5466) confirms the first-order variables tested have only weak individual-level predictive power; they should not be used as a standalone customer-targeting rule.

8. Conclusion

This project set out to test, rather than assume, whether two first-order operational experiences are associated with customer repeat purchase in the Olist marketplace. First-order delivery lateness is confirmed as a real, though modest, statistically significant driver of reduced repeat purchase. First-order freight burden is not confirmed as a driver once the freight metric is corrected for a mechanical confound with order value — a correction this project deliberately surfaced and documented rather than silently carrying forward. Neither effect is strong enough, individually or combined, to reliably predict which specific customers will return, as shown by the modest predictive baseline (ROC-AUC = 0.5466). The recommended path forward is operational, not predictive: prioritise first-order delivery reliability at the population level, allocate resources by exposure and customer value, and validate the causal impact of any intervention through a controlled operational test before committing to scale.

9. Technology Stack and Reproducibility

DuckDB serves as the primary analytical engine (in-process SQL, lazy evaluation) for all cleaning, feature engineering, and aggregation. Statistical testing uses SciPy and statsmodels (chi-square, Mann-Whitney U, proportions z-test, VIF, logistic regression). The predictive baseline uses scikit-learn. Charts use Matplotlib with a consistent neutral-gray/signal-red colour convention tied directly to each chart's underlying significance test, so that colour never implies an effect the statistics do not support. Path resolution is handled by a small config-driven module (`src/config.py`, `config.yaml`) rather than hardcoded paths in each notebook, and a pytest suite (`tests/test_pipeline.py`) checks row-count integrity, absence of duplicate customer identities, and the presence of all expected persisted diagnostic findings.

Appendix A — Repository Structure

01_Proposal/ — problem statement and proposal form

02_Data/ — raw, processed, and analytical data exports, organised by notebook stage

03_Notebooks/ — the six analytical notebooks (01 Data Inventory and Validation through 06 Business Recommendations and Final Report)

04_DuckDB/ — the persistent analytical database (olist_capstone.duckdb)

05_Dashboard/ — live/exported dashboard

06_Report/ — this technical report

07_Presentation/ — presentation deck

src/ — config-driven path resolution module

tests/ — pytest reproducibility checks

config.yaml,

requirements.txt,

README.md — project configuration and documentation